CHAPTER

ATA 75

AIR SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Air System – General

Introduction

The engine air system is designed to control the performance of the engine compressors during different flight regimes so as to prevent engine surging and stalling. It also provides compressed air to aircraft services and for sealing internal engine components.

General Description

Refer Figure 75-1 & *75-2*

The purpose of the Air System is to provide compressed air for:

- Cabin pressurization
- Cooling of hot engine components
- Bearing sealing
- P2.2 bleed valve operation
- P2.7 bleed valve operation
- Reference air pressure for the pressure relief valve
- Scavenging of various bearing cavities
- P3 signal to the FADEC
- Airframe services

Detailed Description

The air system is a critical system for the operation and performance of the engine.

The function of the air system is performed by the:

- Cooling and distribution system
- Compressor control system
- Air indicating system

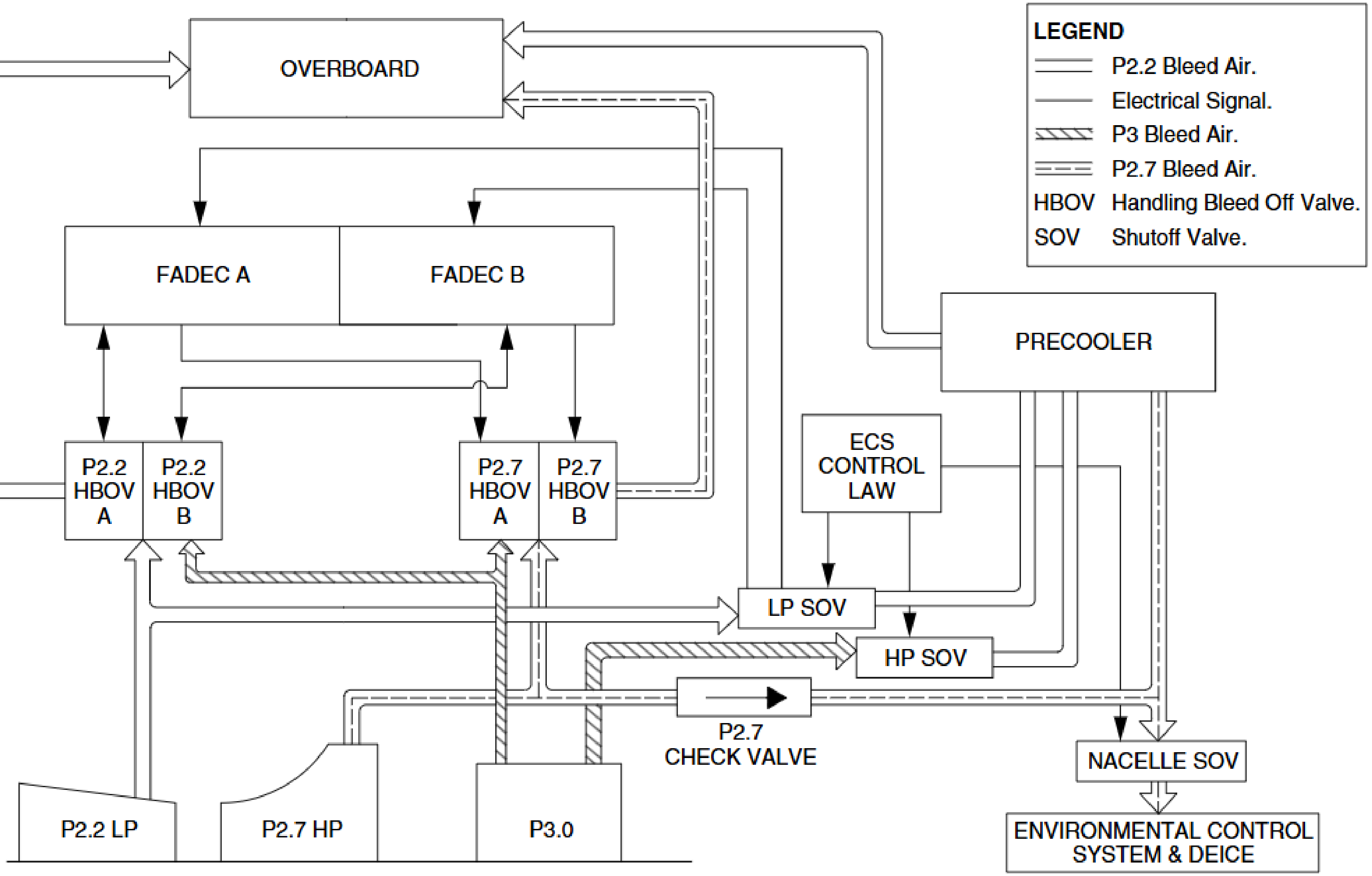

Figure 75-1 Air System General

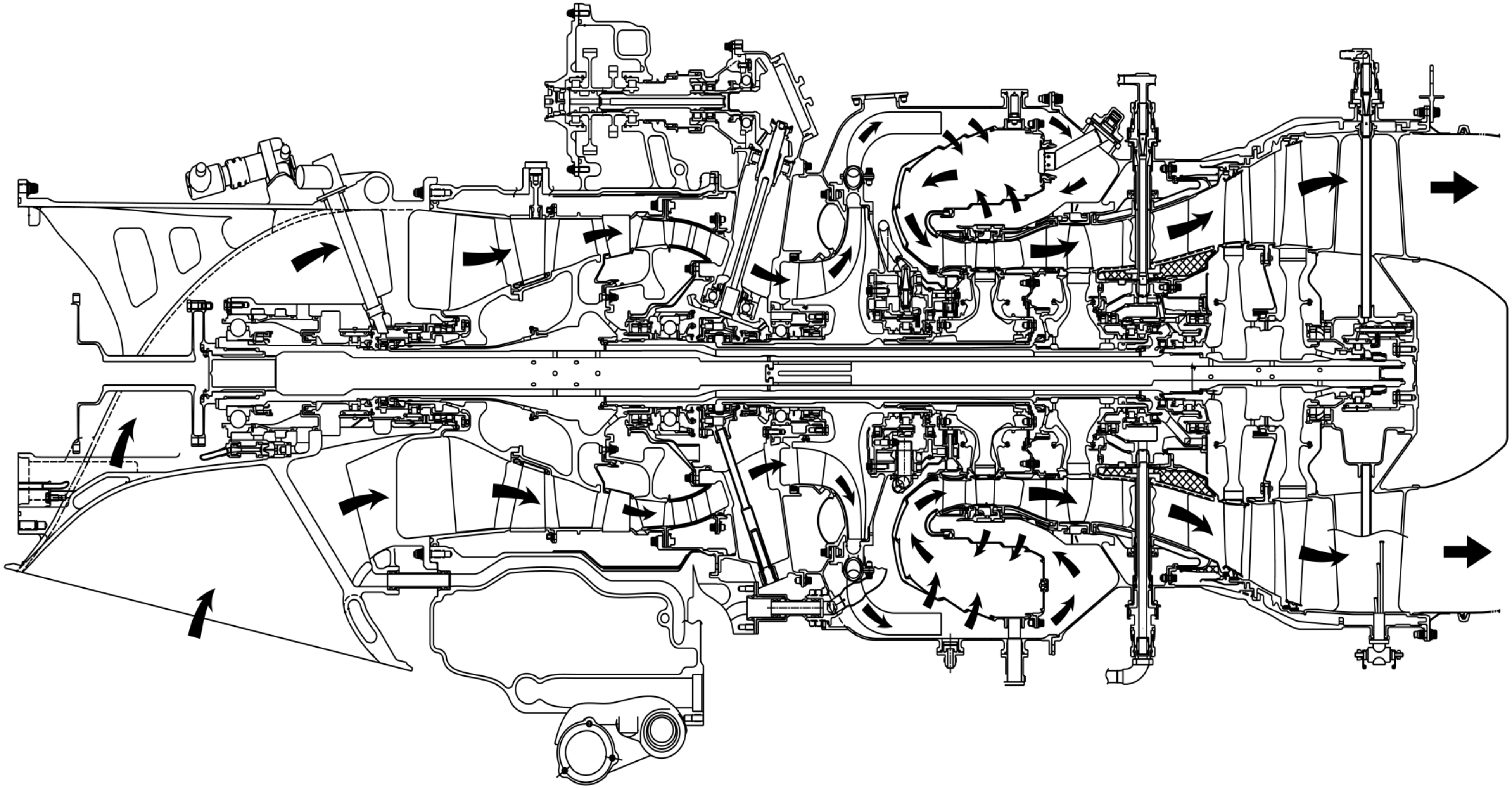

Figure 75-2 Engine Airflow

Cooling and Distribution

Introduction

The air distribution system provides air for cooling of hot section components, bearing oil cavity sealing, scavenging, and aircraft services. The purpose of the Cooling System is to cool the hot engine components. Also, this cooling air is used to scavenge and seal the bearing cavities, for compressor control and by the environmental control system.

General Description

Refer Figure 75-2

The Cooling and Distribution System is a critical system for the operation and performance of the engine. During engine operation air is drawn into the engine and compressed. Some of this air is diverted through internal passages and external tubes. This air is used to cool engine components, seal and scavenge bearing oil cavities and for aircraft services through the environmental control system. Engine cooling is a critical system for the operation and performance of the engine. The engine Cooling System is designed to control the temperature of critical engine components. It also provides air for engine sealing and aircraft services.

The function of the distribution system is performed by the:

- Internal air passages
- Intercompressor Case (ICC) to Turbine Support Case (TSC) cooling air tube

Detailed Description

The engine air distribution system is designed to control the temperature of critical engine components. It also serves to provide air for engine sealing and aircraft services.

Internal Air Passages

Internal air passages direct air flow to seal and scavenge the bearing and seal oil cavities. This air is also used to cool internal engine components.

Refer Figure 75-3

The axial flow compressor is cooled internally using P2.5 air. The P2.5 air leaks into the compressor disc after the last stage of stator blades. From the compressor disc the P2.5 cooling air flows through drilled passages into the turbine shaft and then flows aft to the power turbines. The air also leaks across the carbon seals at No.2 and No. 2.5 bearings to prevent the lubricating oil from leaking away from the bearings.

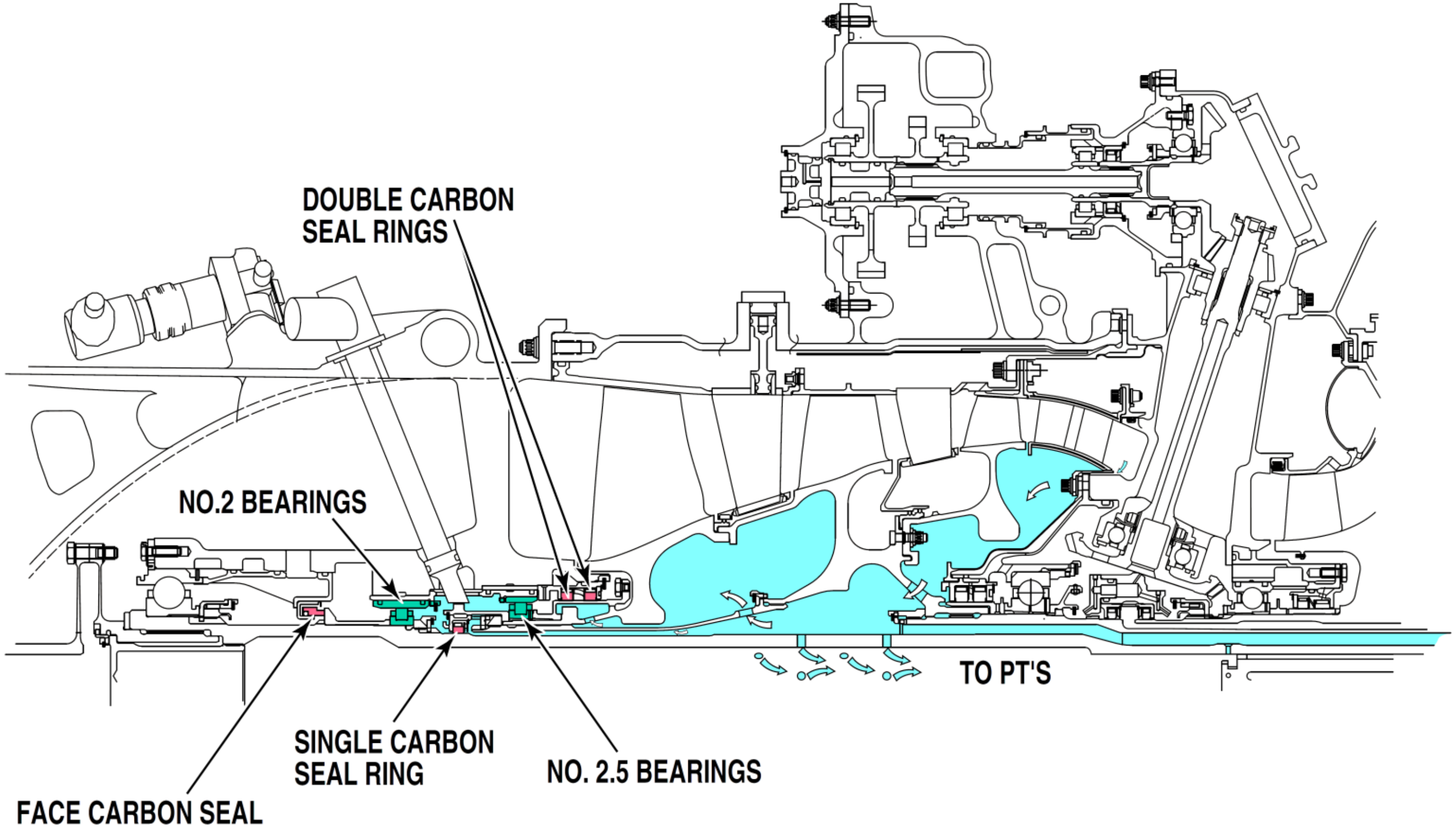

Figure 75-3 Distribution to No. 2 & 2.5 Bearing Cavity

Refer Figure 75-4

The High Pressure (HP) and Low Pressure (LP) turbines are cooled by P3 air flowing in through the blade tips. This P3 air is from the cooling airflow of the combustion chamber liner. P2.5 air flowing through passages from the drive shaft is used to seal the bearing cavities of No. 3, 4 and 5 bearings. P2.5 air is also used to cool the front face of the LP turbine and the rear face of the LP turbine.

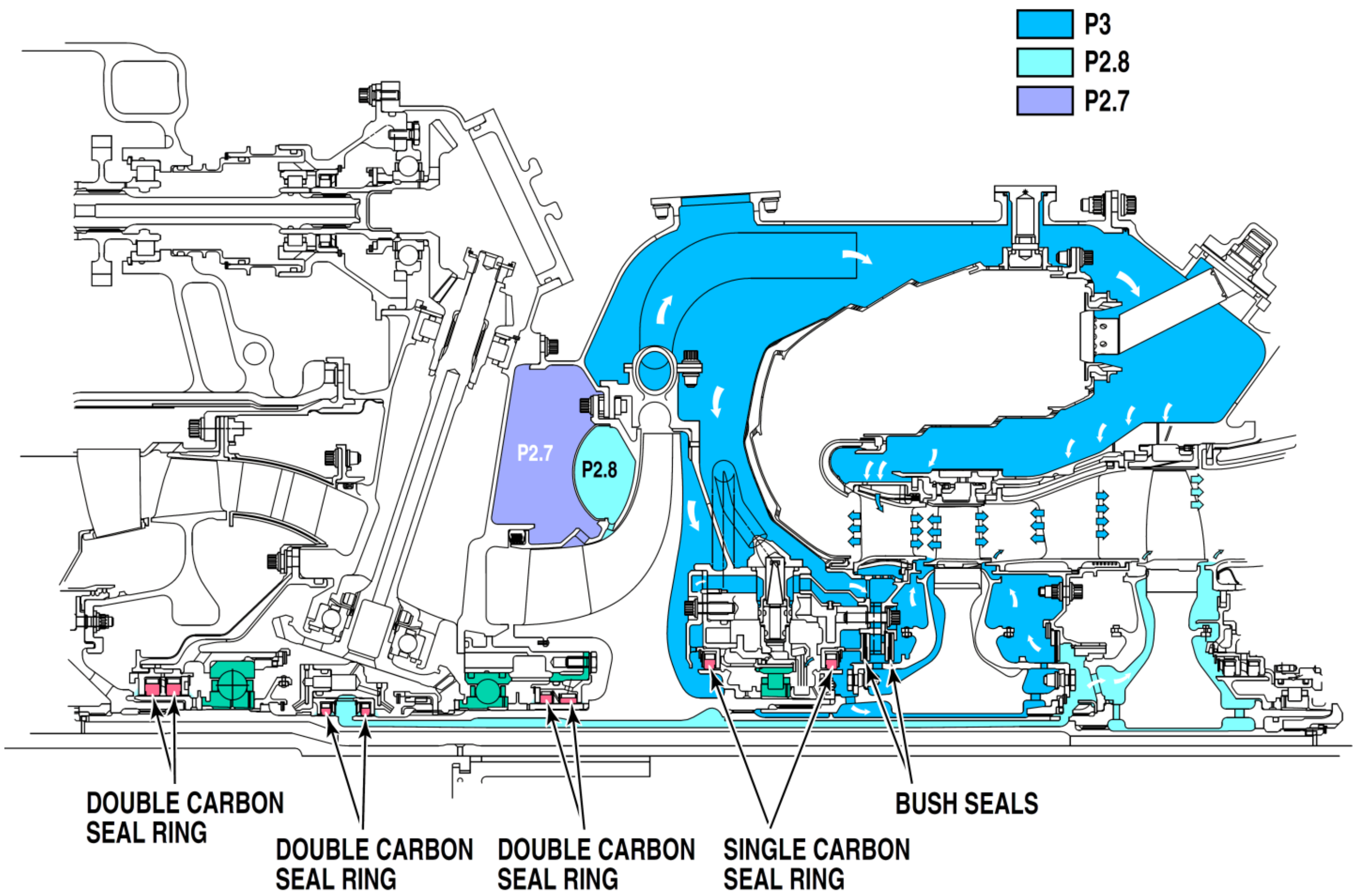

Figure 75-4 No. 3, 4 & 5 Bearing Cavity

Refer Figure 75-5

P2.5 air flowing from the turbine shaft is used to cool the discs and the roots of the blades of the power turbines. P2.7 air is used to cool the power turbine blades internally.

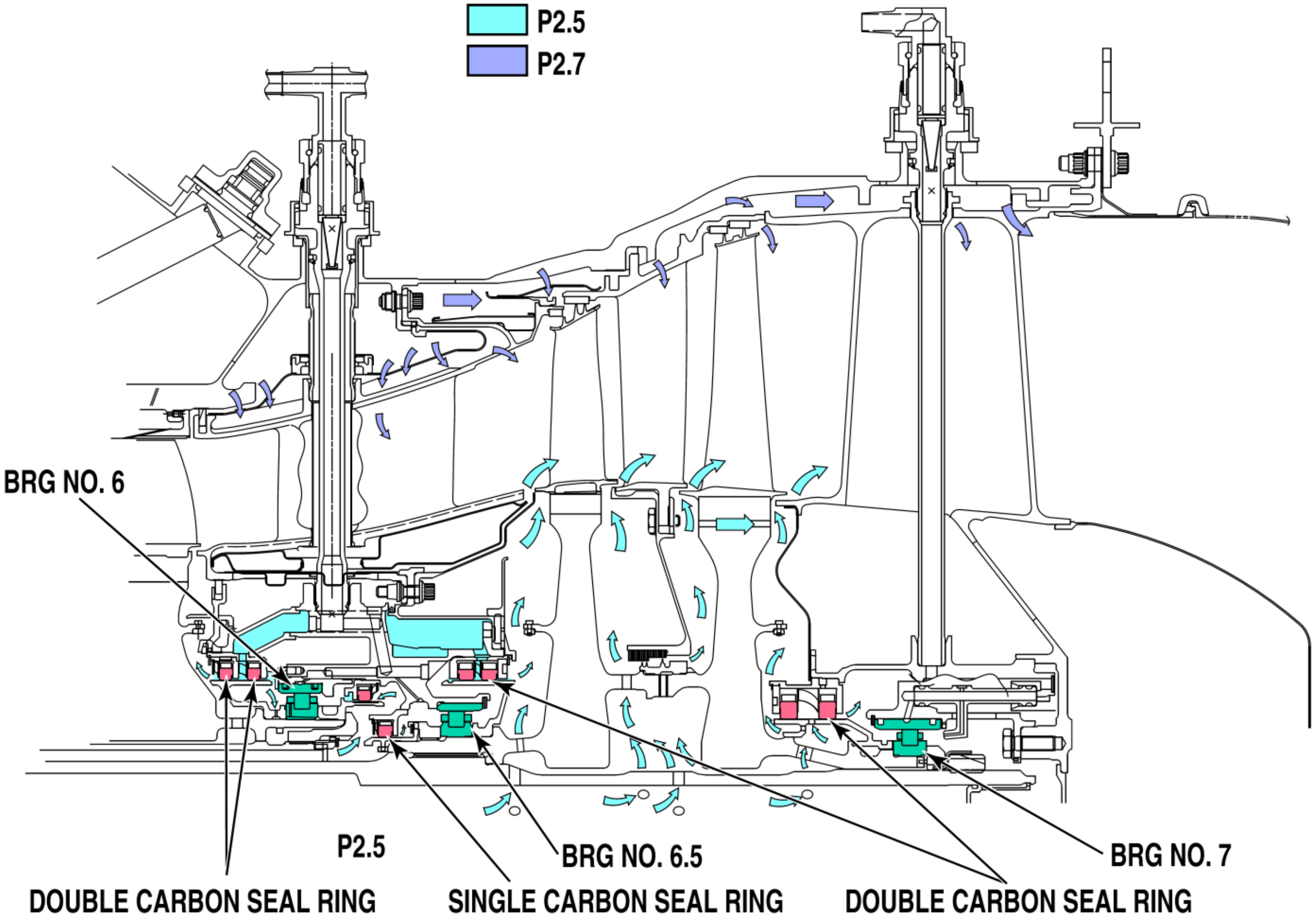

Figure 75-5 No. 6, 6.5 & 7 Bearing Cavity

Intercompressor Case to Turbine Support Case Cooling Air Tube

Refer Figure 75-6

The intercompressor case to turbine support case cooling air tube carries bleed air to the turbine support case to cool the interturbine vane.

If the turbine support case is not cooled, its material properties are adversely affected to the point where it is possible that subsequent failure of a turbine blade may result in uncontained debris.

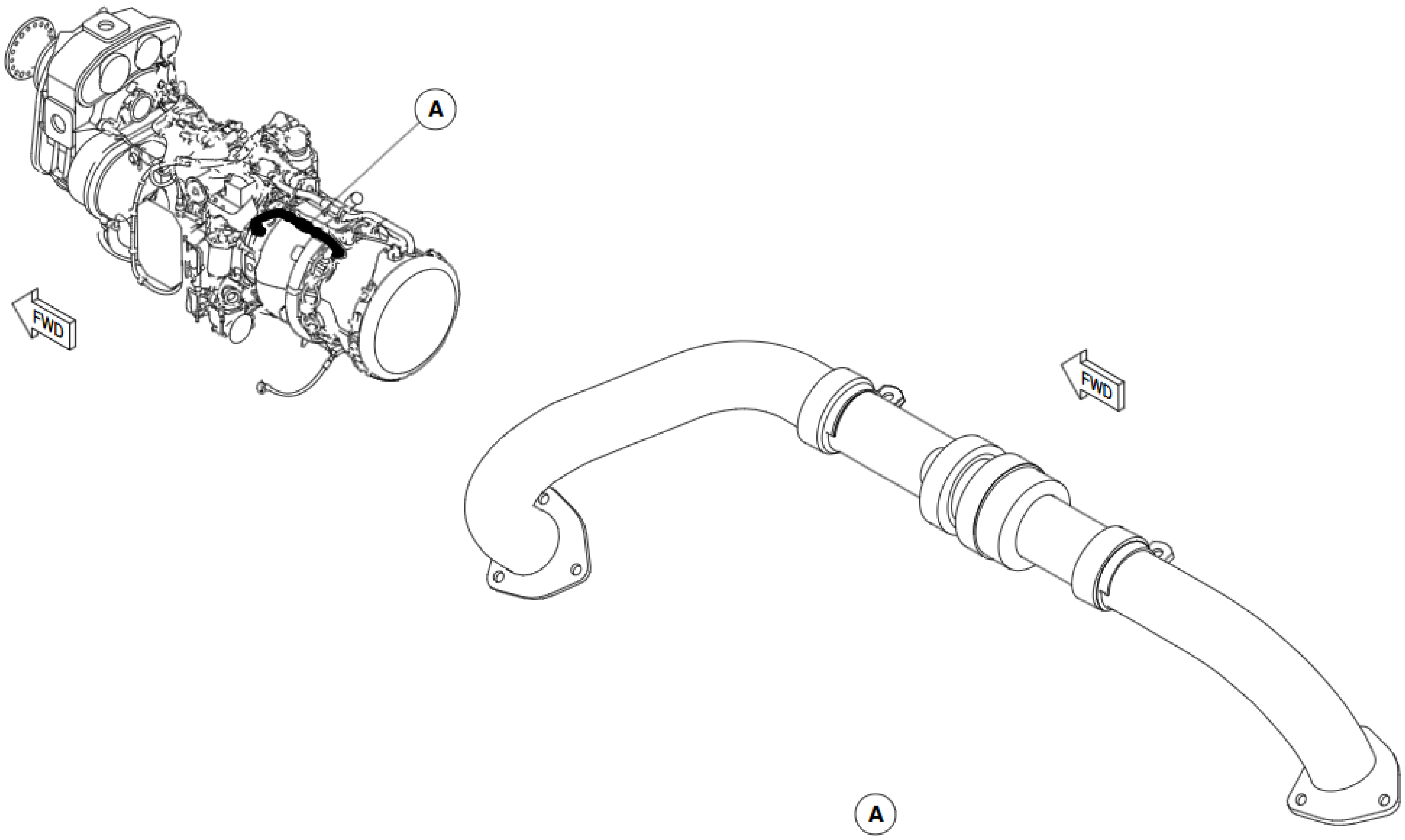

Figure 75-6 Intercompressor Case to Turbine Support Case Cooling Air Tube

Compressor Control

Introduction

The purpose of the Air Compressor Control system is to control the performance of the compressor and to deliver air for aircraft services:

General Description

The function of the Air Compressor Control system is performed by the:

- P2.2 Interstage Bleed Valve
- P2.7 Handling Bleed Valve
- P2.7/P3 Check Valve
- P3 Air Separator
- P3 Air Separator to P2.2 Bleed Valve Air Tube
- P3 Air Separator to P2.7 Bleed Valve Air Tube
- Gas Generator Case to P3 Separator Air Tube

Detailed Description

Refer Figure 75-7

The air compressor control system uses two pneumatic valves for compressor control. They are located on the left and right sides of the engine. They use P3 air for actuation. An electrical signal from the FADEC controls the P3 air.

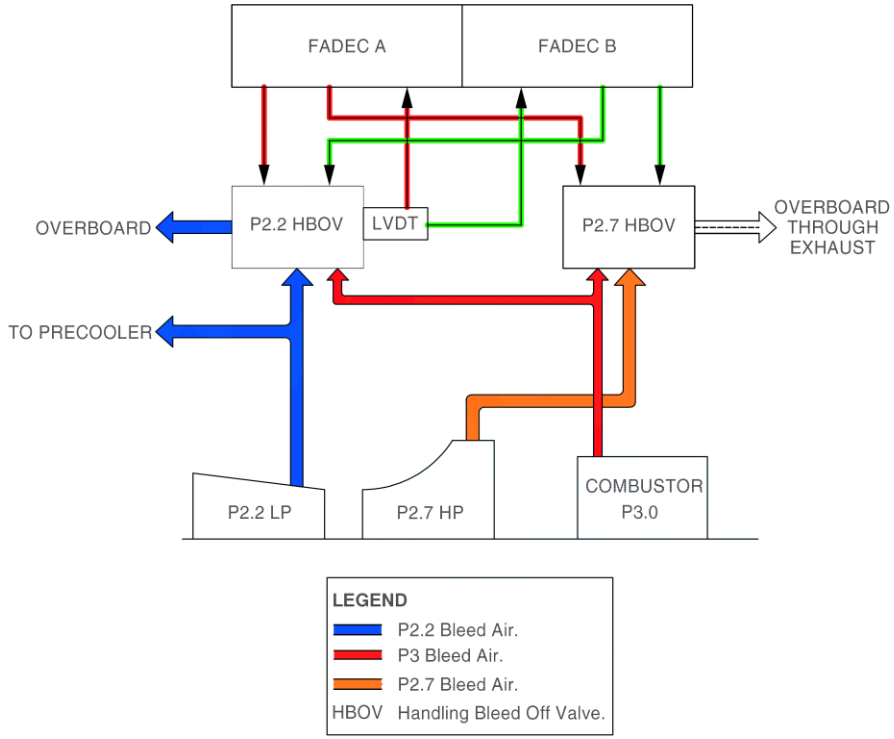

Figure 75-7 Compressor Surge Control System

Compressor Control – Component Description

P2.7 Handling Bleed Valve

Refer Figure 75-8

The P2.7 handling bleed valve is a normally open, dual coil torque motor, on−off in−line valve. It has an inlet, outlet, and flange mounted servo port. There is an electrical connector hard mounted to the torque motor. The P2.7 valve is located on the right side of the intercompressor case.

During engine operation the valve is positioned open or closed by a command signal from the FADEC. The signal is fed to either coil of the torque motor which then closes the servo supply pressure (P3) and increases the vent area causing the control pressure to drop. The compressor bleed pressure (P2.7) pushes on the underside of the valve poppet and opens the valve. When the current to the torque motor is reduced, the vent is closed. This increases the control pressure (P3) and closes the valve. During engine shutdown, a compression spring will open the valve to the full open position.

In the event of a loss of electrical power to the torque motor the valve will fail to the closed position.

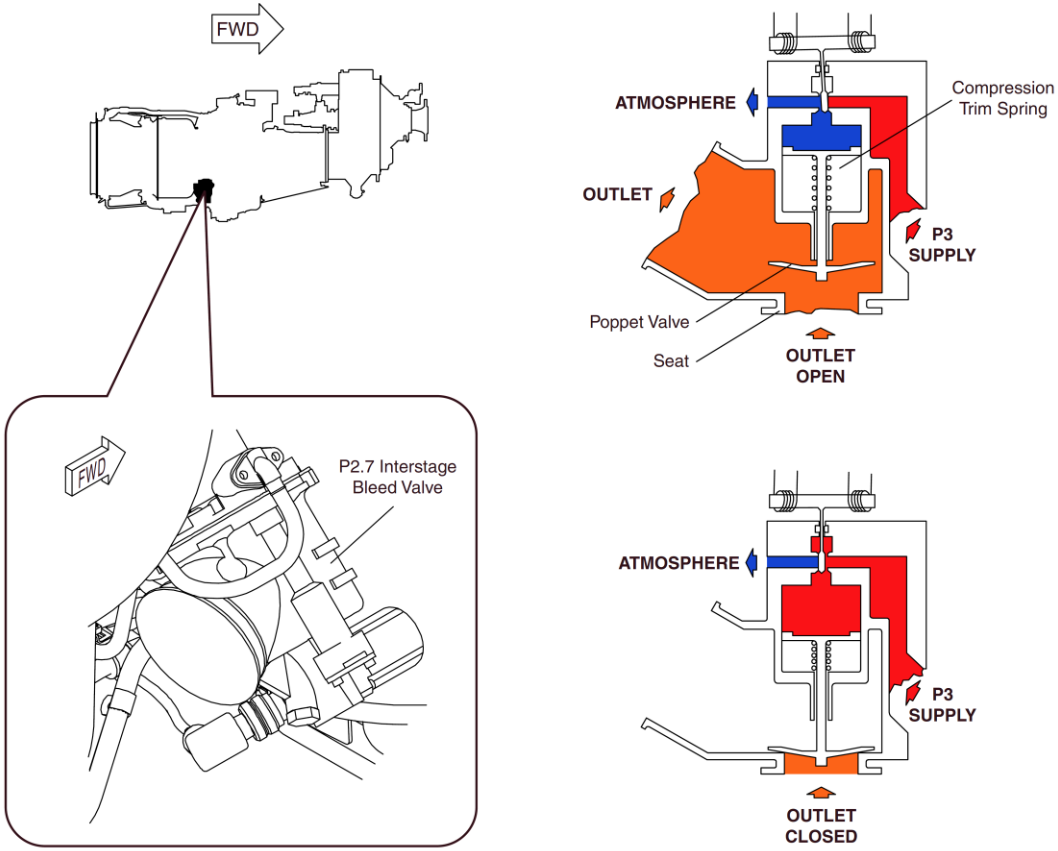

Figure 75-8 P2.7 Handling Bleed Off Valve

P2.2 Interstage Bleed Valve

Refer Figure 75-9

The P2.2 interstage bleed valve is a normally open, dual coil torque motor, modulating in−line valve. It has an inlet, outlet, flange mounted servo port and a Linear Variable Differential Transducer (LVDT) connected to the valve poppet. There is an electrical connector hard mounted to the torque motor. The P2.2 valve is located on the left side of the low pressure compressor case.

During engine start the valve is positioned to maximum bleed. During normal engine operation the valve is modulated in the closing position according to a FADEC schedule. The valve is positioned by a command signal from the FADEC to the valve's torque motor. This signal moves the flapper rod, which changes the control pressure to the valve, upsetting the force balance and moving the valve. When the FADEC receives the LVDT position feedback signal which correlates to its schedule, it commands the torque motor back to the null flapper rod position. This balances the forces on the valve and it remains stationary in its new position until the next FADEC command is received. During engine shutdown the valve is commanded back to the full open position.

The P2.2 valve is designed to provide redundancy in the event of a failure. The torque motor and LVDT each have 2 coils that are handled through 2 separately wired channels. Also, in the event of a power loss to the torque motor, its null bias will cause the valve to open to its full bleed position. The valve is also designed to fully open if there is a loss of servo supply pressure.

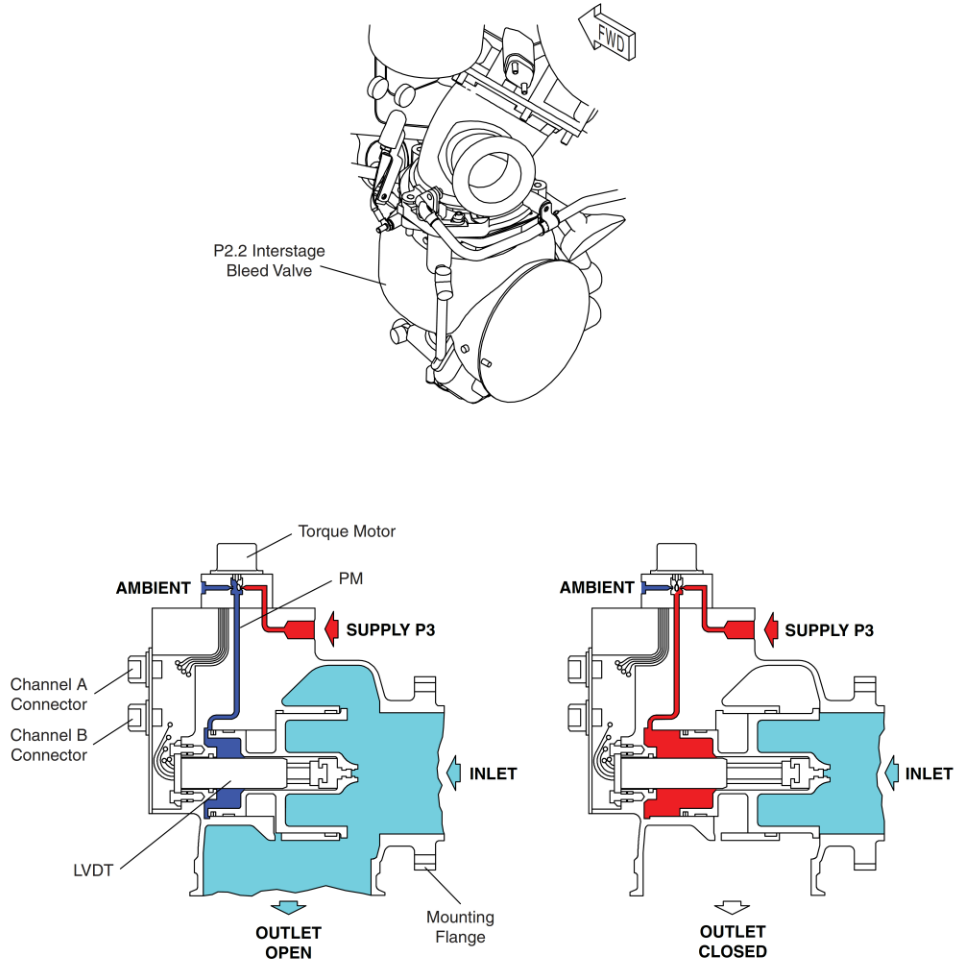

Figure 75-9 P2.2 Interstage Bleed Valve

Refer Figure 75-10

The P2.2 valve is designed to provide redundancy in the event of a failure. The torque motor and LVDT each have 2 coils that are handled through 2 separately wired channels. In the event of a power loss to the torque motor, its null bias will cause the valve to open to its full bleed position. The valve is also designed to fully open if there is a loss of servo supply pressure.

The P2.2 HBOV is fully modulating. It is controlled by the FADEC to provide surge margin during steady state operation.

The amount of valve opening is biased closed by 50% whenever the ECS is demanding P2.2 bleed for precooling P3.

This is done by transmitting a discrete signal from the ECS ECU to the ECIU and then to the FADEC on an ARINC 429 data bus.

For a transient surge the amount of opening is biased open by 50% for the duration of the surge.

For a steady state surge the amount of opening is biased open by 14%.

The FADEC will retain this bias until a FADEC Fault Clear is done. If further surging is detected the FADEC will bias the HBOV open by an additional 14% to a maximum of 70%.

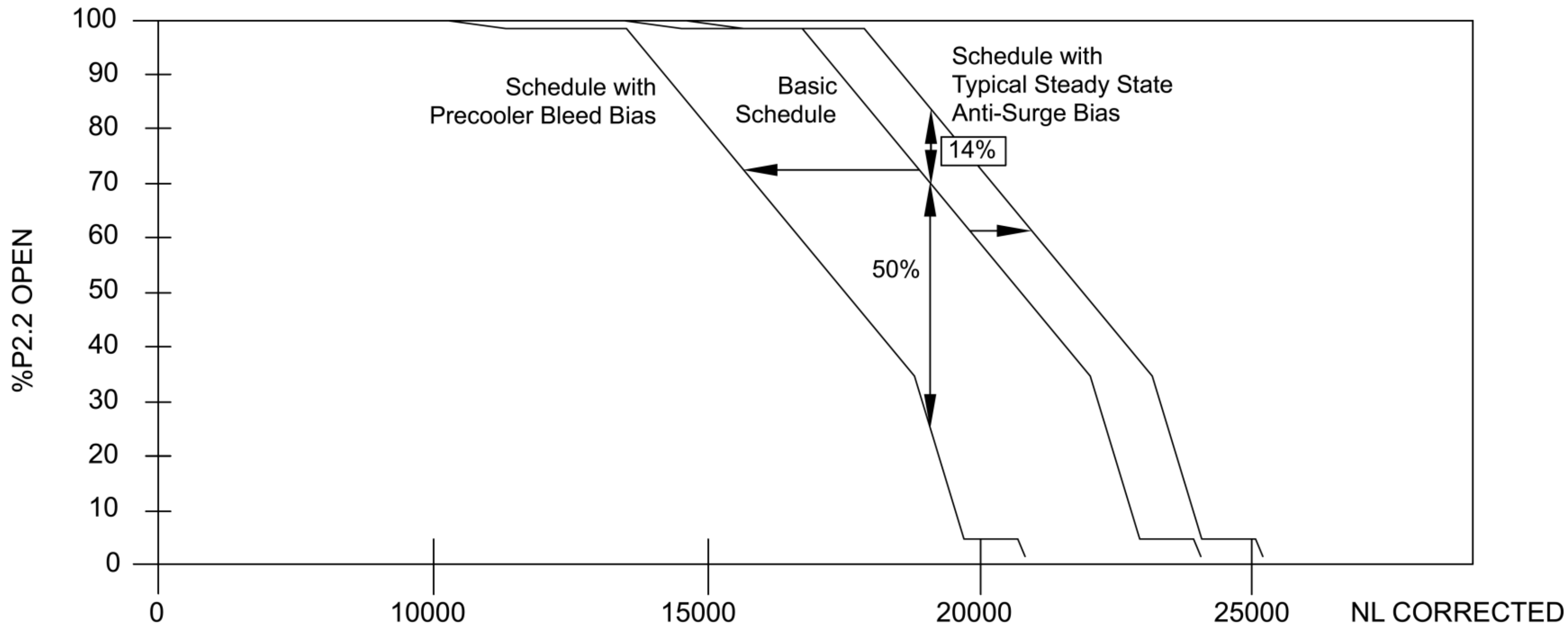

Figure 75-10 P2.2 Interstage Bleed Valve Steady State Schedule

P2.7/P3 Check Valve

[Refer Figure 75-11](#)

The P2.7/P3 check valve is part of the environmental control system. The environmental control system uses either P2.7 or P3 air as supply. If the P3 air is being used, the P2.7/P3 check valve will prevent backflow into the high pressure compressor.

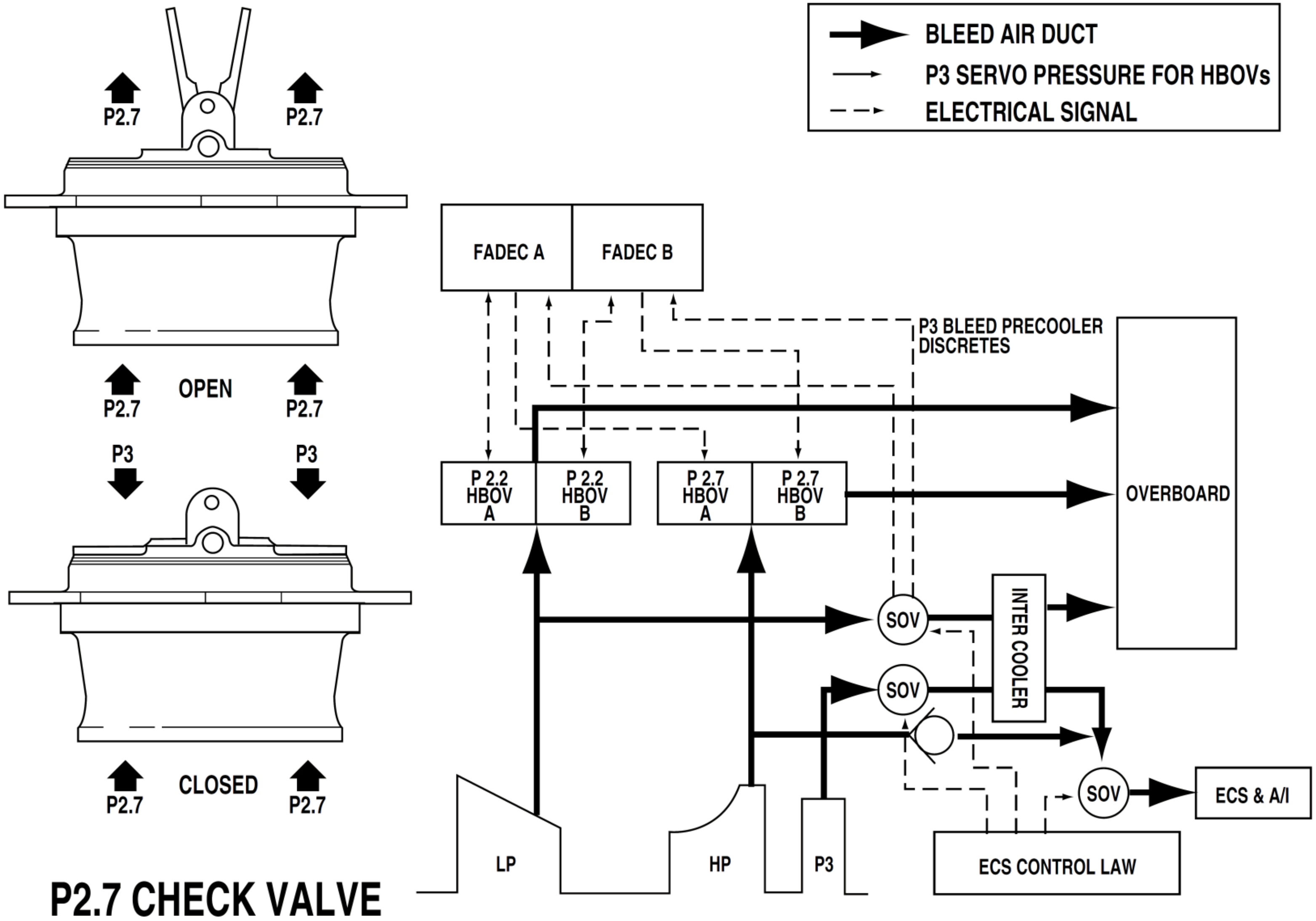

Figure 75-11 P2.7/P3 Check Valve

P3 Air Separator

Refer Figure 75-12

In order to prevent contamination damage of the P3 servo inlet port, a centrifugal separator is used to remove dirt and dust from the P3 air.

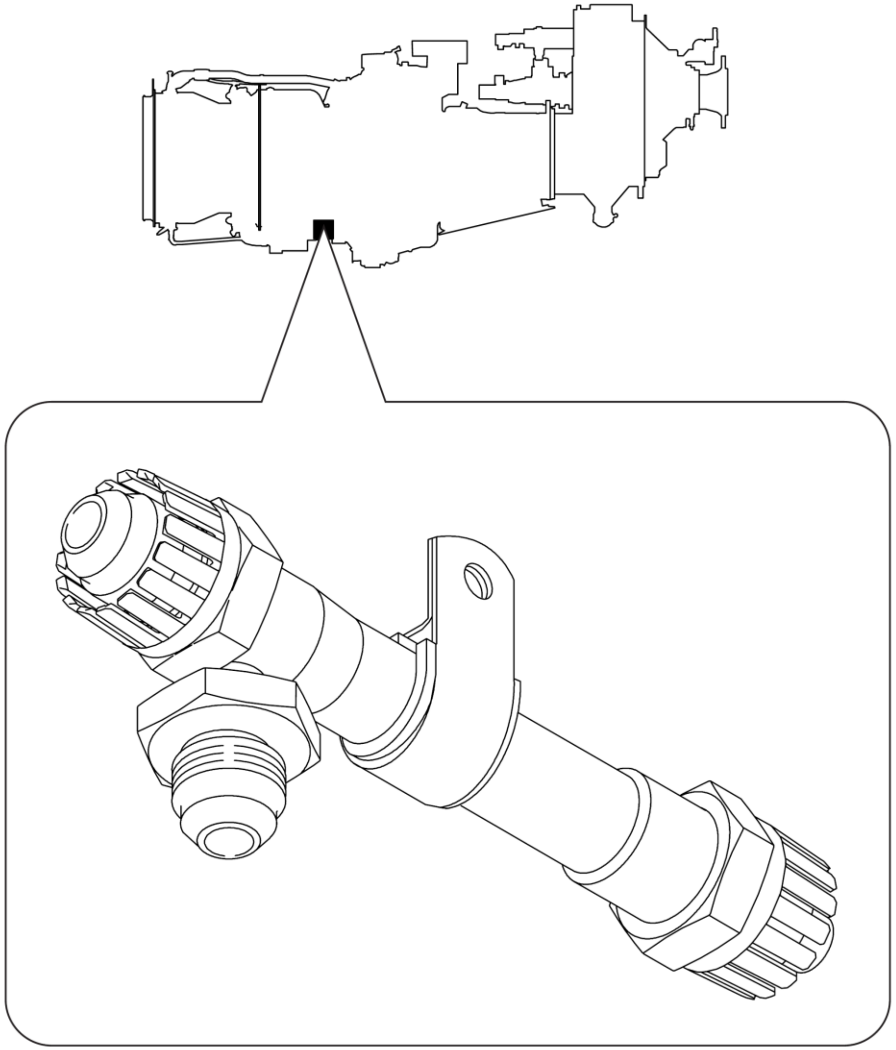

Figure 75-12 P3 Air Separator

P3 Air Separator to P2.2 Bleed Valve Air Tube

Refer Figure 75-13

This tube supplies P3 air to the P2.2 bleed valve torque motor servo port. This air is used to modulate the position of the P2.2 bleed valve.

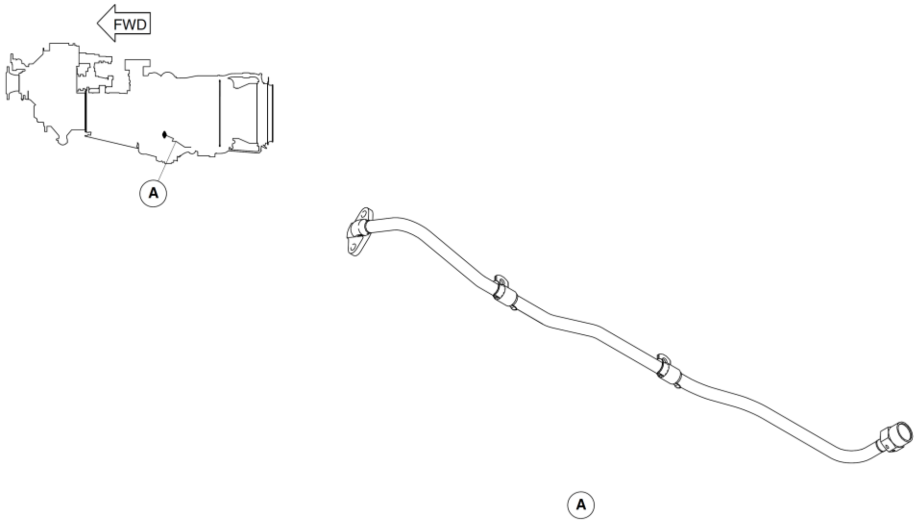

Figure 75-13 P3 Air Separator to P2.2 Bleed Valve Air Tube

P3 Air Separator to P2.7 Bleed Valve Air Tube

Refer Figure 75-14

This tube supplies P3 air to the P2.7 bleed valve torque motor servo port. This air is used to modulate the position of the P2.7 bleed valve.

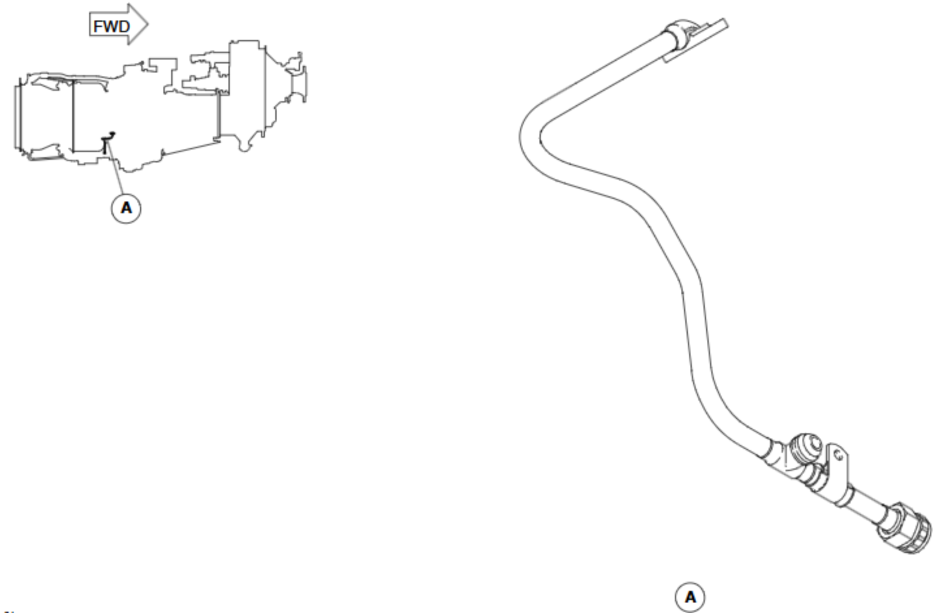

Figure 75-14 P3 Air Separator to P2.7 Bleed Valve Air Tube

Gas Generator Case to P3 Separator Air Tube

Refer Figure 75-15

This tube carries P3 air from the gas generator case to the P3 separator.

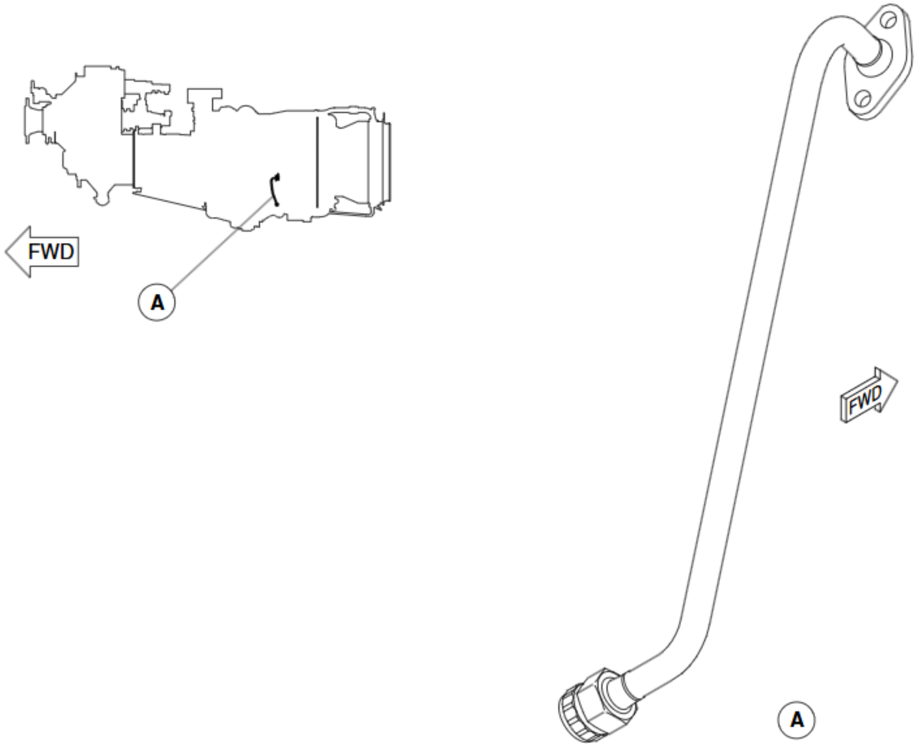

Figure 75-15 Gas Generator Case to P3 Separator Air Tube

Compressor Control – Operation

Refer Figure 75-16

During engine START and operation at low N_L the P2.2 handling bleed off valve is held OPEN by command from FADEC EEC.

The HBOV torque motor and LVDT receive signals through the following pins:
Torque motor signals are from FADEC to the HBOV
LVDT signals are to and from FADECEEC to the HBOV.

The signals from the N_L sensor go to:

- FADEC EEC Channel A.

The signals from the N_L sensor go to:

- FADEC EEC Channel B.

HBOV Basic Schedule

As the engine accelerates the signal from the N_L sensor will increase. When the signal equals approximately 70% corrected N_L, FADEC EEC will begin to modulate P2.2 HBOV in a closing direction, by controlling the torque motor of the valve.

Pre-cooler Bleed Bias Schedule

If the P2.2 SOV in the bleed air system is OPEN, a signal from a switch on the valve will cause FADEC EEC to modulate the P2.2 HBOV closed by 50% of its normal opening schedule.

P2.7 HBOV

After engine start, with no compressor stall/ surge detected, FADEC EEC will signal the P2.7 HBOV to close.

The HBOV will remain closed through the remainder of engine operation, unless FADEC EEC detects that the P2.2 HBOV is not preventing a compressor stall/surge. In this situation, FADEC EEC will command the P2.7 valve to OPEN.

Compressor Stall/Surge Detected

If compressor stall surge is detected, FADEC EEC will open the HBOV's, decrease the fuel flow and energize both spark ignitors.

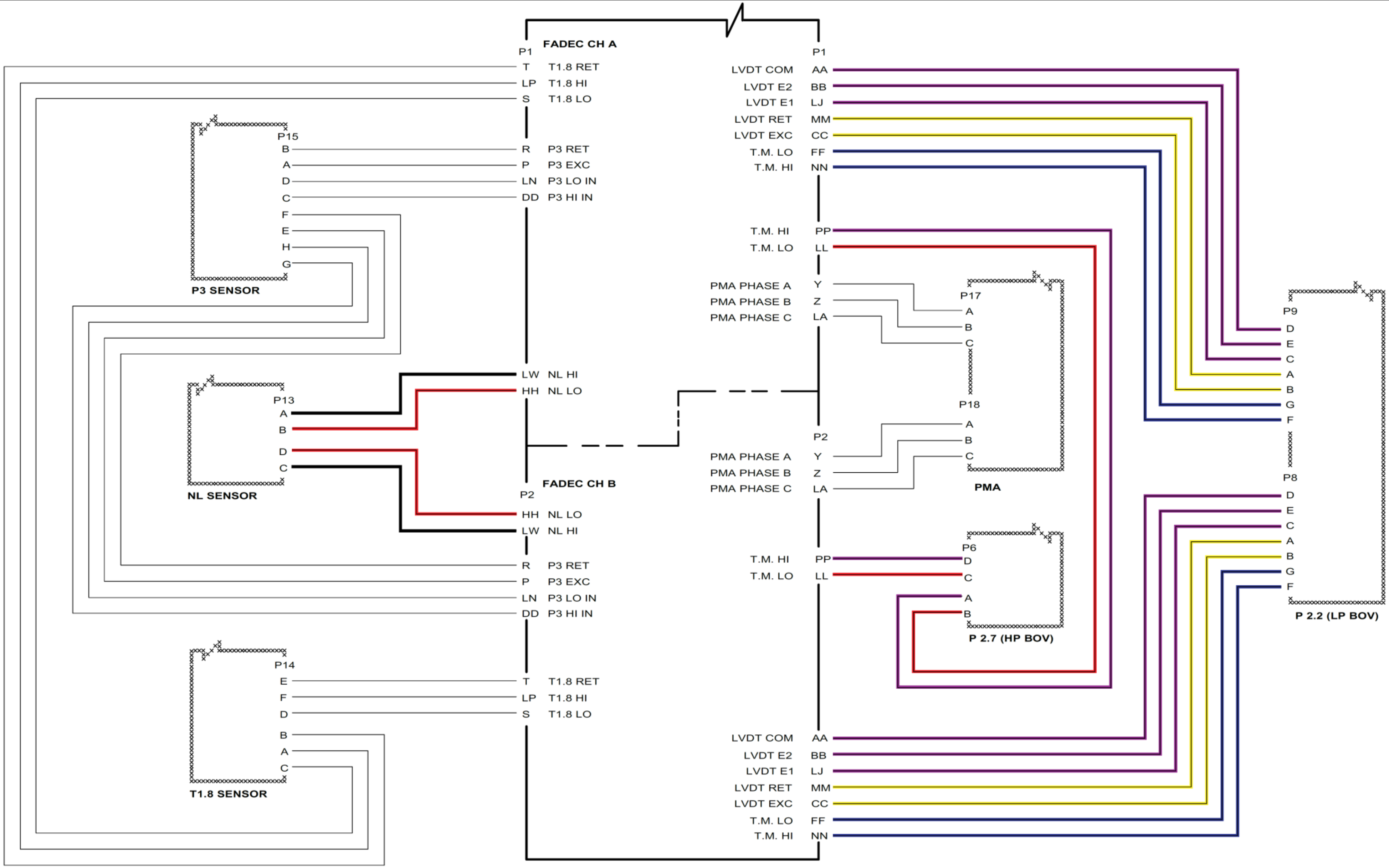

Figure 75-16 HBOV's Controls

Air Indication

Introduction

The Air Indication system senses compressor discharge pressure at the exit from the HP compressor and transmits this pressure, as an electrical signal, to the related FADEC. This signal is used for engine control.

General Description

Refer Figure 75-17

A transducer receives P3 air pressure and converts this pressure to a proportional electrical signal that is then sent to the FADEC. The FADEC uses this signal to control the bleed valves.

The function of the Air Indication system is performed by the:

- P3 Pressure Transducer
- P3 Transducer to Gas Generator Case Air Tube

Detailed Description

The Air Indication system uses a transducer to convert P3 pressure to an electrical signal that the FADEC uses to control bleed valve operation.

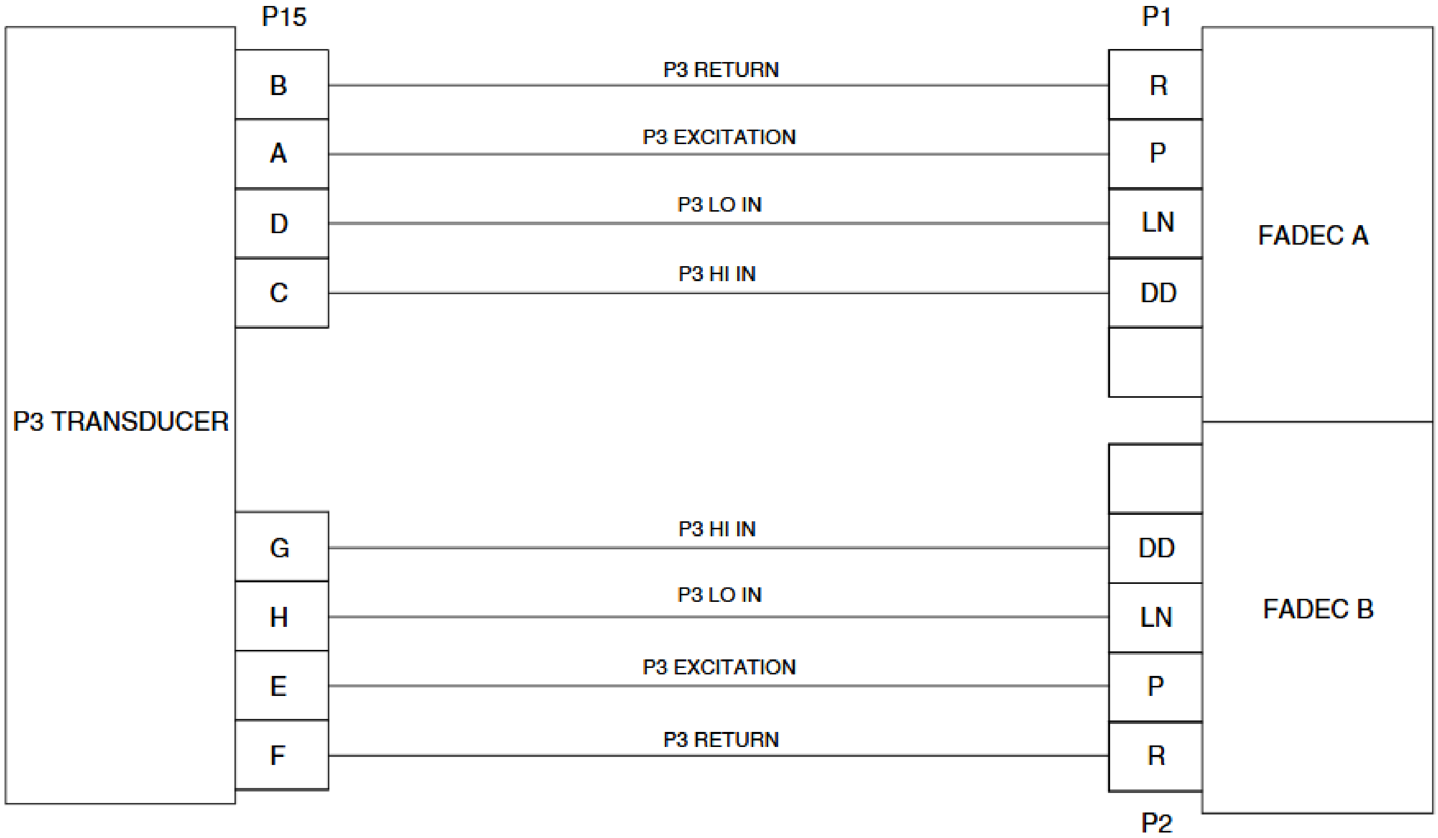

Figure 75-17 Air Indication System Schematic

Air Indication – Component Description

P3 Pressure Transducer

Refer Figure 75-18

The transducer is located adjacent to the rear of the accessory gearbox and is connected to the gas generator case by a stainless steel tube and to the FADEC through an electrical wiring harness.

The P3 transducer senses compressor discharge pressure at the exit from the HP compressor. The P3 transducer converts this pressure into a proportional electrical signal and sends it to the FADEC. The FADEC uses this signal to control the bleed valves.

The P3 transducer is a 2 channel device and in the event one channel fails the other channel can fully provide the signal to the FADEC.

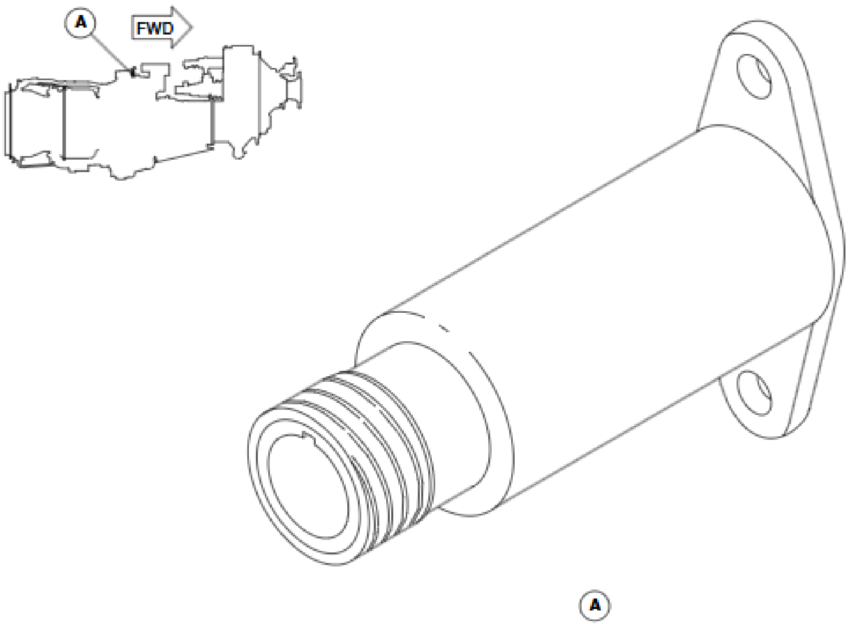

Figure 75-18 P3 Pressure Transducer

P3 Transducer to Gas Generator Case Air Tube

Refer Figure 75-19

This tube supplies P3 air to the P3 transducer. This air supply is used by the FADEC for engine control.

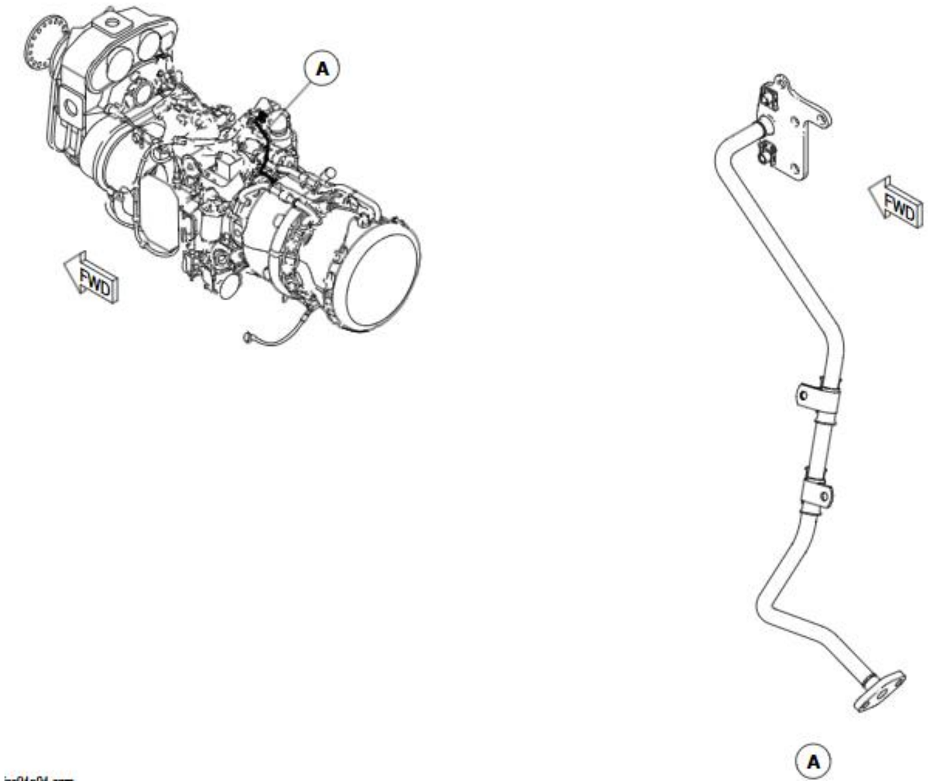

Figure 75-19 P3 Transducer to Gas Generator Case Air Tube